

Lecture 34

① $A \rightarrow 3 \times n$ matrix

$$Ax = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{no solution}$$

$$Ax = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{exactly one solution}$$

9) What can we tell about m, n, r ?

$$\Rightarrow m = 3$$

No nullspace, so
no column without
pivots.

$$\boxed{n = r}$$

$$\boxed{n = 1, 2 \quad m = 3}$$

ii) True / False

a) $\det(A^T A) = \det(A A^T)$

b) $A^T A$ is invertible.

c) $A A^T$ is positive definite.

~~a)~~ True $\Rightarrow$ This is actually false as only for square.

b) $A^T A \rightarrow n \times n$ matrix with same rank

Since $r = n$

$\text{rank}(A^T A) = n$

Non, invertible. ✓ True.

c) $A A^T \rightarrow m \times n \rightarrow 3 \times 3$ symmetric

$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
False

$\begin{bmatrix} 0 & 1 & 0 \end{bmatrix} =$

$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
positive semi-definite

iii) Prove $A^T y = c$ has at least one solution for every c in fact infinitely many solutions.

$$\Rightarrow A^T \rightarrow n \times 3$$

$$\text{rank} = n < 3$$

has a null space

rows = rank = full row rank.

So, definitely at least one solution exists. and since null space exists then infinitely many solutions exist.

$$(2) \quad A = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}$$

a) Solve $Ax = v_1 - v_2 + v_3$,

$$\Rightarrow x = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

b) Suppose $v_1 - v_2 + v_3 = 0$ then give reason for whether this is the only combination for which it is true.

$\Rightarrow v_1 - v_2 + v_3 = 0$ tells us that

$\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ is in the nullspace of A .

then all multiples $c \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ are also in nullspace so not unique.

c) Suppose v_1, v_2, v_3 are orthonormal
then what combination of v_1 & v_2
is closest to v_3 .

$$\Rightarrow \|v_3 - av_1 - bv_2\| \rightarrow \min.$$

$$= \|v_3\|^2 + a^2 \|v_1\|^2 + b^2 \|v_2\|^2$$

$$\boxed{a = b = 0}$$

③ $A = \begin{bmatrix} .2 & .4 & .8 \\ .4 & .2 & .8 \\ .4 & .4 & .4 \end{bmatrix}$

a) what are its λ 's?

$\lambda = 1$ since markov matrix.

$$\lambda_1 + \lambda_2 + 1 = 0.8$$

$$\lambda_1 + \lambda_2 = -0.2$$

Since $\text{col } 1 + \text{col } 2 = 2 \cdot \text{col } 3$

then $\lambda = 0$ is also an eigenvector.

$$\lambda = -0.2, 0, 1$$

b) $u_R = A^R u_0 \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

what does this approach?

$$\Rightarrow U_R = C_1 \lambda_1^n x_1 + \dots$$

$$\lambda = 0 \quad \begin{bmatrix} 2 & 4 & 3 \\ 0 & -6 & -3 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

$$\lambda = 1 \quad \begin{bmatrix} -0.8 & .4 & .3 \\ & -0.6 & \textcircled{0.18} \\ & & \times \times \end{bmatrix} \quad \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix}$$

0.45

$$\lambda = -0.2 \quad \begin{bmatrix} .4 & \textcircled{.4} & .3 \\ 0 & 0 & 0 \\ 0 & 0 & .3 \end{bmatrix} \quad \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$U_R = C_2 (1)^k \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix} + C_3 (-0.2)^k \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$u_0 = \cdot \quad c_2 x_2 = c_2 \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix}$$

$$u_0 = c_2(1) \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix} + c_3(1) \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

$$c_1 \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \\ 0 \end{bmatrix}$$

$$\# \quad c_1 = 2c_2$$

$$\# \quad c_1 + 3c_2 + c_3 = 0$$

$$5c_2 + c_3 = 0$$

$$\boxed{c_3 = -5c_2}$$

$$\# \quad c_1 + 3c_2 - c_3 = 10$$

$$2c_2 + 3c_2 + 5c_2 = 10$$

$$\boxed{c_2 = 1}$$

4) 9) Project onto line $a = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$

$$\Rightarrow P = \frac{aa^T}{a^T a} \Rightarrow \frac{\begin{bmatrix} 4 \\ -3 \end{bmatrix} \begin{bmatrix} 4 & -3 \end{bmatrix}}{25}$$

$$= \frac{1}{25} \begin{bmatrix} 16 & -12 \\ -12 & 9 \end{bmatrix}$$

ii) $\lambda = 0$, $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ singular

$$\lambda = 3, \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

What is the matrix??

$$\Rightarrow \begin{bmatrix} -2b & b \\ -2\lambda b & \lambda b \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 4 & -2 \\ 2 & -1 \end{bmatrix}$$

$$\boxed{b = -2}$$

Also, can be done by

$$A = S \Lambda S^{-1}$$

iii) $A \neq B^T B$ for any B .

$$B \rightarrow n \times 2$$

$B^T B \rightarrow$ symmetric matrix for sure

All non-symmetric matrix A ?

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

iv) A matrix that has orthogonal eigen vectors but is not symmetric.

⇒ # Symmetric matrices definitely have orthogonal eigen vectors

Skew-symmetric also have orthogonal eigenvectors with complex eigenvalues

Orthogonal matrices also have orthogonal eigen vectors

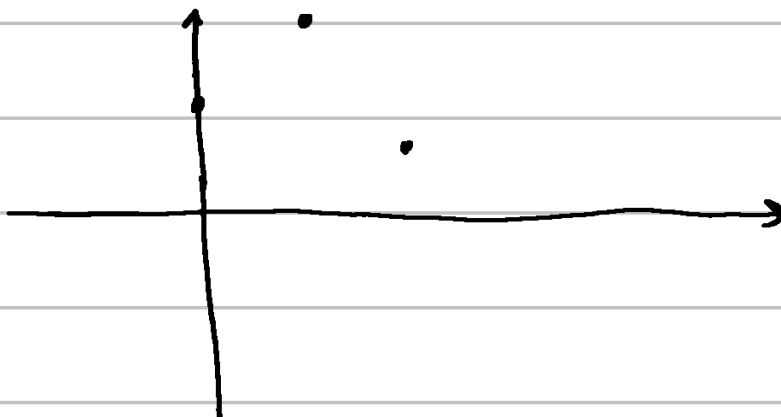
5)
$$A \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} b$$

$$\begin{bmatrix} \hat{c} \\ \hat{d} \end{bmatrix} = \begin{bmatrix} 11/3 \\ -1 \end{bmatrix}$$

i) What is projection p of b onto A .

$\Rightarrow$ it is $\begin{bmatrix} \hat{c} \\ \hat{d} \end{bmatrix}$

ii) Draw the least line problem that corresponds to this system.



iii) Find a different vector b for which the least squares solution would be zero.

$$\rightarrow \begin{bmatrix} \hat{c} \\ \hat{d} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{when } b \text{ is orthogonal to col } A$$

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$b_1 + b_2 + b_3 = 0$$

$$b_2 + 2 = 0$$

$$b_2 = -2$$

$$\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$b_1 - 2 + 1 = 0$$

$$b_1 = 1$$